TIANDA SUN

PhD Candidate in Computer Science | LLM Reasoning, Agentic Tool Use & Interpretability

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Professional Summary

Final-year Computer Science PhD working on LLM reasoning and agentic tool use, with research in evaluation, RL post-training, and mechanistic interpretability. Two first-author papers published at EMNLP 2026: one Main Conference paper and one Findings paper. Built a GRPO tool-use training pipeline and KinshipQA, a controlled reasoning benchmark; current submissions examine tool-call representations and the limits of residual interventions. Experienced in reproducible multi-seed and cross-model experiments.

Research Interests

LLM Reasoning • Agentic Tool Use • Benchmark Design & Evaluation • RL Post-training • Mechanistic Interpretability • Knowledge-Graph Reasoning

Education

Ph.D. in Computer Science — University of York, UK Sep 2022 – Present

- Thesis: *Knowledge Graph-Guided Reasoning in Large Language Models: Towards Interpretable Question Answering*
- Focus: multi-hop reasoning, knowledge graph integration, agentic tool use, benchmark development
- Supervisor: Dr. Dimitar Kazakov

M.Res. in Computer Science — University of York, UK Sep 2021 – Sep 2022

- Thesis: Relation Extraction in Financial Reports using Distant Supervision with BERT and PCNN

M.Sc. in Advanced Computer Science — University of Liverpool, UK 2017 – 2018

B.Sc. in Software Engineering — Tianjin Polytechnic University, China 2011 – 2015

Selected Conference Papers

Tianda Sun, Dimitar Kazakov. *"Peak-Then-Collapse and the Four Interface Channels of Knowledge-Graph Tool Use."*
EMNLP 2026 Main Conference | ACL Anthology | arXiv | Code

Found a tool-grounded answer rate rising from 3.8% to 9.6% before collapsing to 0% within 50 steps, replicated across 4 seeds and 7 reward designs; one-iteration self-distillation reached 40.0% exact match.

Tianda Sun, Dimitar Kazakov. *"Kinship Data Benchmark for Multi-hop Reasoning."*
Findings of EMNLP 2026 | ACL Anthology | arXiv | Code

Built a procedurally generated benchmark spanning 7 kinship systems and up to 6 reasoning hops; evaluated 6 LLMs and used controlled interventions to investigate failures in cultural-rule application.

Preprints & Manuscripts Under Review

Tianda Sun, Dimitar Kazakov. *"Tool-Call Dependency Structure is Linearly Decodable in LLM Agent Residual Streams."*
arXiv preprint, 2026; under review | arXiv:2605.25310 | Code

Probed tool-call dependency representations using linear probes, positional and random-label controls, and activation patching across multiple benchmarks and 2 model families; distinguished representation evidence from behavioural control.

Tianda Sun, Dimitar Kazakov. *"From Target Selection to Condition Reuse: Auditing Residual Interventions in Tool-Use Models."*
arXiv preprint, 2026; under review | arXiv:2610.31686 | Code

Audited residual edits in Qwen3.5-27B across 32 fresh worlds, showing that familiar target-selection gains do not establish condition reuse, independent eligibility judgments, or complete-action correctness.

Additional Publications

Tianda Sun, Dimitar Kazakov. *"KGEIR: Knowledge Graph-Enhanced Iterative Reasoning for Multi-Hop Question Answering."* **RANLP 2025 Workshop (R2LM)**. ACL Anthology | Code

Jiawei Chen, Zhihao Chen, Tianda Sun, Kun Liu, et al. *"TOGGLE Delineates Fate and Function within Individual Cell Types via Single Cell Transcriptomics."* **bioRxiv**, 2025. DOI

Tianda Sun, Jamie Carr, Dimitar Kazakov. *"A Hybrid Question Answering Model with Ontological Integration for Environmental Information."* **DAO-XAI 2024 @ ECAI** (workshop). PDF

Nunung Nurul Qomariyah, Tianda Sun, Dimitar Kazakov. *"NLP Analysis of COVID-19 Radiology Reports in Indonesian using IndoBERT."* **IEEE IBIOMED 2022** (co-author; British Council Newton Fund)

Research Experience

PhD Research: Knowledge Graph-Guided LLM Reasoning — University of York 2022 – Present

- Built a single-GPU GRPO post-training pipeline (verl + vLLM) for knowledge-graph tool use; used unit-tested rewards and multi-seed experiments to isolate sparse interface feedback as the bottleneck behind train-time collapse.
- Probed tool-call dependency structure with linear edge probes, random-label and positional controls, counterfactual tests, and layer-wise activation patching across 2 model families and multiple interactive benchmarks.
- Audited residual interventions using independent readouts of target selection, eligibility judgments, and complete actions; released an archived example and a CPU workflow for reproducing numerical paper assets.
- Created KinshipQA with a synthetic RDF/OWL + SPARQL pipeline, controllable 1–6 hop complexity, and a cross-model protocol over 6 LLMs; identified cultural-prior interference as a distinct reasoning failure mode.
- Developed KGEIR, integrating dynamic knowledge-graph construction with iterative LLM reasoning and improving multi-hop F1 by ~4.5% over a Tree-of-Thought baseline.
- Released reproducible open-source research artifacts for benchmark generation, evaluation, RL post-training, and interpretability analysis.

M.Res. Research: Financial Relation Extraction — University of York 2021 – 2022

- Implemented a distant-supervision approach using the FIBO ontology for automatic training-data generation.
- Compared BERT and PCNN architectures for relation extraction, analyzing transformer vs. CNN performance.

Technical Skills

Programming: Python (NumPy, Pandas, SciPy), Java, SQL, Bash, C++ (basic)
ML / RL: PyTorch, HuggingFace Transformers, scikit-learn, GRPO / PPO, LoRA / DoRA, verl, vLLM
Evaluation & Interpretability: benchmark design, multi-seed / cross-model evaluation, linear probing, activation patching, bootstrap confidence intervals
NLP & Knowledge Graphs: SPARQL, RDF/OWL, Neo4j, NLTK, spaCy, OpenNRE, LangChain
LLM APIs: OpenAI, Anthropic Claude, Google Gemini; Ollama
Infrastructure: Linux, Git, Docker, HPC / Slurm clusters, Jupyter, VS Code
Languages: English (fluent), Mandarin (native)

Open Source Contributions

- **KinshipQA** and **KGEIR**: benchmark-generation, evaluation, and knowledge-graph reasoning code with reproducible experiments.
- **KG-GRPO**: RL post-training; **Tool-Call Dependency Probe** and **Residual Condition Reuse**: representation analysis and intervention audits.

Teaching Experience

Graduate Teaching Assistant — University of York 2022 – 2025

- DEEP: Deep Learning and Evolution of AI Paradigms — practical sessions on neural network implementation.
- EVAC: Evaluating and Validating AI Claims — critical analysis of AI research methodology.